Limitations and Safety of the Project Code

This project contains very small databases: only 3 books, 1 robot, 6 shelves. It has not been tested for multi-agent systems and its functionality in such situations cannot be accounted for. Since this is purely computational and does involve real libraries or robots, we have to simulate the shelves and sensors so these are only approximations in that sense. The LLM can always misinterpret or ignore instructions if the prompt is not strict enough. The way that it is coded, the robot will stop physically checking a shelf once it is "trusted", while this saves battery, it leads to incorrect data being reported to the user. Despite attempts at fixing these issues, the frontend cannot fully support NLP inputs, the book-id always has to be mentioned, attempts at trying to do this have broken the frontend. You must take care to keep your API key private; running the program costs real money.